Large Language Model-Based Explainable Anomaly Detection Using Prompt Engineering in Operational Data*

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Abstract

Reliable anomaly detection is critical in Industrial Control System because malfunctions can lead to physical damage and human casualties. To address the lack of explainability in existing machine learning models, we propose an explainable method leveraging prompt engineering and Large Language Model. This method transforms multivariate time-series data into textual context to extract natural language rules that capture complex sensor-actuator relationships. By providing detection results alongside violated rules and logical rationales, this approach significantly enhances the transparency and trustworthiness of the system.

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1 Introduction and Proposed Approach

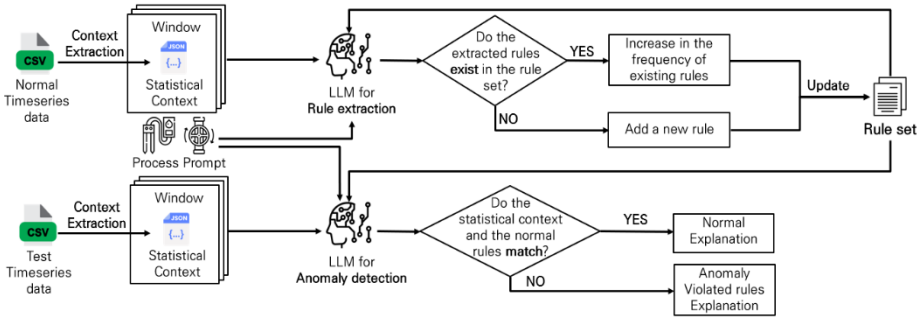


Figure 1 Workflow of Explainable Anomaly Detection based on LLM Using Prompt Engineering for ICS

Industrial Control System (ICS) can cause severe physical damage or even loss of life when anomalies occur. But traditional machine learning–based does not explain the basis for detection, making it difficult for operators to respond immediately [1].

To address this limitation, we propose a large language model (LLM)-based anomaly detection approach for ICS that extracts natural language rules describing normal sensor–actuator correlations and explicitly explains the causes of detected anomalies, as illustrated in Figure 1. The proposed approach consists of three stages. First, multivariate timeseries data are transformed into context windows. Sensor values are summarized as increasing, decreasing, stable, fluctuating based on z-score normalized slope and standard deviation. Actuator states are summarized through run-length encoding, such as ON(30s)->OFF(70s), to preserve state duration and switching order. Second, normal contexts are provided to the LLM together with process descriptions and device metadata. The LLM generates human-readable IF-THEN rules that describe normal relationships between sensors and actuators. Third, the system prompt includes the process overview and the learned rule set. The LLM then returns a structured output that includes the anomaly status, the violated rules, and the reasons for judging those rules as violated.

2 Experiments and Results

The experiments use the Secure Water Treatment (SWaT) dataset, a realistic water-treatment testbed with six process stages and 51 sensor/actuator signals sampled every second [2]. This study focuses on the P1 process, which stores raw water in tank T101, and considers five variables: FIT101, LIT101, MV101, P101, and P102. Normal data were used to construct rule extraction windows, while attack data were applied for anomaly detection. To ensure feasibility in a single-GPU environment, we employed Llama-3.1-8B with 8 billion parameters, without fine-tuning.

From 495,000 normal time steps, 4,949 100-second windows were generated, yielding 46 normal-operation rules. Eight P1-targeted SWaT attack scenarios were evaluated, including forced MV101 opening, abnormal operation or stopping of P101/P102, and LIT101 manipulation. The proposed method achieved a precision of 0.800, recall of 0.816, and F1-score of 0.808. Compared with LSTM-ED (P: 0.989, R: 0.688, F1: 0.812) and GDN (P: 0.978, R: 0.685, F1: 0.806)[3], it shows comparable recall despite lower precision, while providing operator-level explainability through the window index, raw data range, decision, confidence, violated rule, explanation, and LLM input context.

3 Conclusion

This poster proposed an explainable ICS anomaly detection method using LLM and prompt engineering, which automatically derives human-interpretable sensor-actuator rules and provides explicit rule-violation evidence upon detection. Future work will extend the analysis beyond P1 to multi-stage process interactions and evaluate whether cross-process rules can improve explanations for complex attacks.

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